Smaller separating norms and a signed finalizer for Diophantine ZFC consistency

Research synthesis prepared for the ProveIt project

23 October 2026

Abstract

We compare and combine the eight current Diophantine consistency packages in ProveIt, numbered 16–23, at the fixed repository revision recorded below. Two new algebraic reductions give concrete improvements. First, a polynomial in twelve strictly positive integer unknowns has exact degree 1081 and a division-free circuit of 116 binary arithmetic operations. Its three-equation presentation has maximum degree 1072 and 112 operations. This improves the former twelve-variable degree minimum 1145 while matching the former operation minimum 116. The key change is the uniform bound $|y'| < (T^2 + XI^2)\sqrt{G}$ for every relevant algebraic conjugate, together with a smaller coefficient for the first Pell root.

Second, a concrete Jones specialization in nine nonnegative unknowns has exact degree $93264\,5^{33} + 288$, compared with $161844\,5^{33} + 484$ in package 18, and uses 20 547 operations instead of 20 531. This is a 42.374% degree reduction at a cost of sixteen operations. It combines structured radical bounds with a signed divisibility finalizer that avoids squaring the divisor and dividend.

Both constructions use the complete, already published 943-tile coefficient tuple of package 19, with $B = 2^{5^{10}}$. We rebuild its four actual integers and include their compressed binary representations. The nine-variable index constants are defined by bounded finite recipes, rather than expanded. Composing the new algebra with the repository's ordinary compiler and K215 calibration theorems proves in ZFC that each stated unsolvability assertion is equivalent to Con(ZFC). We give detailed proofs, exact polynomial circuits, independent algebraic checks, and a reproducible archive. We claim improvements within the inspected family, without claiming global minimality or a new end-to-end proof-assistant certificate.

Repository snapshot. `ae3007ca4467011e0e051c84acb4a1dd13971b0e`. All comparisons use that snapshot; the historical package numbers are retained for attribution.

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1 The objective, its logical meaning, and the comparison

1.1 What is being constructed

Write $\mathbb{N} = \{0, 1, 2, \ldots\}$. Fix an effective standard presentation of ZFC, and let Con(ZFC) mean that there is no finite derivation of falsity in that presentation. We construct fixed integer-coefficient polynomials P_{12} and F_9, with the following meanings:

$$\mathrm{ZFC} \vdash \left(\nexists \boldsymbol{x} \in \mathbb{Z}_{>0}^{12} : P_{12}(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con(ZFC)}, \tag{1}$$

$$\mathrm{ZFC} \vdash \left(\nexists \boldsymbol{v} \in \mathbb{N}^{9} : F_9(\boldsymbol{v}) = 0\right) \longleftrightarrow \mathrm{Con(ZFC)}. \tag{2}$$

The coefficients are independent of whether ZFC is consistent. None is chosen by waiting for a refutation or a halting computation. Every displayed auxiliary letter in the polynomial definitions denotes a substituted polynomial expression; it is not an extra quantified unknown.

The domains in (1) and (2) are part of the claims. In particular, the twelve-variable formula must not simply be interpreted over unrestricted integers. The nine-variable formula can be converted to nine positive unknowns by substituting $v_j = x_j - 1$; this preserves exact degree but adds nine subtraction gates to the supplied circuit.

1.2 A precise meaning of simpler

For an explicitly represented polynomial or system, at least five measures matter: the number and domain of unknowns, total degree, the number of equations, the number of arithmetic operations in a shared straight-line circuit, and the description or bit size of its coefficients. These measures compete. Replacing a system by a sum of squares removes equations without removing unknowns and generally doubles the maximum degree. Replacing several square roots by a primitive element removes unknowns but increases degree. Fixed integers are free inputs in the circuit model, although computing or printing those integers may be expensive.

Our operation model counts every binary $+$, $-$ and $\times$, including multiplication by a literal integer and squaring. Reuse of a previously computed value is free. Comparisons in a system are free. All fixed powers in the nine-variable expression are expanded into multiplication gates. The cost of constructing a fixed coefficient is recorded separately.

There is no presently justified single global optimum to report. The concrete achievements are two improved points in this collection of size tradeoffs. In particular, neither result reduces the existing nine-variable count, and the earlier 67-operation equation and low-degree lifted systems retain advantages on their own measures.

1.3 Comparison with the current repository

The following data summarize the current positive twelve-variable products; the baseline entries are from the pinned catalogue and the individual packages [1, 2, 3, 4, 5, 6, 7, 8].

Construction	Degree	Operations	Main method
17-primitive12	2601	167	primitive element
18-jones9, direct form	1609	151	radical combination
19-wang943, direct form	1257	139	saturated radical reduction
20-root12	1145	123	nested root separation
21-norm12	1353	116	efficient norm circuit
22-monic12	1529	117	monic polynomial merger
23-separating12	1369	144	separating coefficients
This report	**1081**	**116**	smaller uniform weights

Package 16 contributes a thirteen-positive-variable form of degree 515 at 84 operations, and a separate unrestricted-integer reduction; it is not a twelve-variable competitor. The new product improves the previous minimum twelve-variable degree by 64, or approximately 5.59%, and reduces the degree of the former 116-operation product by 272, or approximately 20.10%. The new three-equation system retains the best stated count of 112 operations while reducing its maximum degree to 1072.

For nine natural unknowns, put $\mathcal{L} = 5^{33}$:

Construction	Exact degree	Operations
18, improved majorants	$161844\mathcal{L} + 484$	20 531
This report	$93264\mathcal{L} + 288$	20 547

The variable count is unchanged. The degree falls by approximately 42.374138%, while the circuit grows by sixteen operations, about 0.078%. The exact new degree is

$$10\,857\,358\,574\,867\,248\,535\,156\,250\,288.$$

The complete base coefficient tuple is taken from package 19 rather than splicing the coefficients of different compilers. This reuse is a synthesis of existing work, not a newly discovered coefficient-size record.

1.4 The proof chain and what is inherited

The literal machine is K215, with initial state 0, an all-zero two-sided tape, and halt state 215. The repository proves its ordinary calibration [10]:

$$\mathrm{Halt}(K215, \mathrm{blank}) \longleftrightarrow \text{there exists a finite ZFC refutation.}$$

The current Lean endpoint named `KEndpoint215.blank_halts_iff_zfc_refutation` and the corresponding Rocq endpoint provide separate formal evidence for the machine theorem. We inspected the current Lean endpoint but did not rebuild the full Lean/Rocq developments. The mathematical ZFC-provability claim uses the ordinary proof, not an inference from the soundness of Lean.

Package 19 supplies the marked periodic Wang-tile language and the entire fixed coefficient tuple. Its interface theorem identifies the resulting native mask constraints with a blank halt. The inherited Pell arithmetic recovers those constraints from its equations; conversely it constructs canonical positive witnesses from a halting tableau. The signed main-root and signed auxiliary-quotient extensions are required when roots are eliminated. The proofs retained in `vendor/inherited/`

record those extensions. The new twelve-variable argument changes only the root-combining part of that arithmetic proof.

The nine-variable construction starts from the concrete 32-positive-variable quadratic system already specialized in package 18. Its sum of squares is a quartic. Jones's compression and its repository implementation give an equivalent condition with eight outer natural variables and four arithmetic tests [12, 3, 11]. We replace the final polynomial combining those tests and its root bounds. Both new steps receive complete proofs below.

Thus the proof structure is a composition of stated theorems, with a clear boundary between reused compiler mathematics, new algebra, and finite computations. A checksum confirms which integer or machine table was used; it does not prove the unbounded compiler theorem. A circuit identity confirms an algebraic transformation; it does not establish consistency.

1.5 Consequences and limitations of the equivalences

The conclusions are representation theorems, not proofs of either consistency or inconsistency. Under the usual consistency hypothesis for ZFC, Gödel's second incompleteness theorem prevents ZFC from proving the unsolvability side of (1) or (2). Consistency alone should not be used to assert that ZFC cannot prove the opposite, existential statement; an additional arithmetic soundness hypothesis is needed for that stronger two-sided assertion. Our construction does not require either hypothesis to prove the displayed biconditionals inside ZFC.

2 A degree-1081 equation in twelve positive unknowns

2.1 The polynomial, with no hidden unknowns

The twelve supplied integers are

$$(J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, r) \in \mathbb{Z}_{>0}^{12}.$$

Let (β, K_0, M_C, M_F) be the fixed tuple of Section 4. In particular, M_F is the shifted mask. Define the common polynomial expressions

$$\begin{aligned}
&q = \beta J + 1, && X = wq, && Y = sq^3, && E = XY,\\
&k = \eta + \zeta, && c = kY + \eta, && a = E + Y, && H = 4a + 3,\\
&\Delta = a^2 + H = (a+2)^2 - 1, && \ell = EkY, && A = 1 + \ell(\ell + k), && G = 1 + \Delta c^2.
\end{aligned}$$

The packed rank and two outer expressions are

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
C_w &= q - F - \alpha,\\
N_k &= k - hE - R, \qquad N_t = (K_0 + w)C_w + q - F - z(q-1).
\end{aligned} \tag{3}$$

The normalized strong and auxiliary quantities are

$$\begin{aligned}
&I = ic^2, && D = 1 + \Delta I^2, && Q = \Delta^2 I^2, && d = Q - 1,\\
&v = cT, && b = c + RD, && e = Q(Dv^2 + b^2) - 1, && B_a = 2Qvb.
\end{aligned}$$

Our replacements for the separating weights are

$$\boxed{W = T^2 + XI^2, \quad L = 1 + HW, \quad J_\tau = 4HW, \quad S = X + ac + Hr.} \tag{4}$$

For a polynomial in a formal letter u, first reduce it modulo $u^2 - g$ to $p_0 + p_1 u$, and define

$$\mathrm{Norm}_{u^2=g}(p) = p_0^2 - g p_1^2.$$

This is a finite operation on integer polynomials, not a new function in the Diophantine language. Define

$$N = \mathrm{Norm}_{u^2=G}\,\mathrm{Norm}_{\tau^2=A}\left(\left[d(S - Lu - J_\tau\tau)^2 - H^2 e\right]^2 - H^4 B_a^2 D\right). \tag{5}$$

Then the three presentations are

$$\mathcal{S}_{12}: \quad N_k = 1, \quad N_t = 1, \quad N = 0; \tag{6}$$

$$P_{12} = \boxed{N_k N_t (1 + N) - 1;} \tag{7}$$

$$Q_{12} = (N_k - 1)^2 + (N_t - 1)^2 + N^2. \tag{8}$$

2.2 An entirely elementary expansion of the norm

For readers who prefer a formula containing only addition and multiplication, the following expressions expand (5). They also specify a short shared evaluator. The letter A_0 below is unrelated to A:

$$\begin{aligned}
P_0 &= QDv^2, & U_0 &= Qb^2, & e &= P_0 + U_0 - 1,\\
V_0 &= S^2 + L^2 G, & V_1 &= 2SL, & A_0 &= J_\tau^2 A,\\
C_0 &= d(V_0 + A_0) - H^2 e, & C_1 &= dV_1,\\
K_f &= 4d^2 A_0, & Z &= 4H^4 P_0 U_0,\\
C_{s0} &= C_0^2 + G C_1^2, & C_{s1} &= 2C_0 C_1,\\
W_0 &= C_{s0} - K_f V_0 - Z, & W_1 &= C_{s1} - K_f V_1,\\
N_0 &= W_0^2 + G W_1^2 - 4Z K_f V_0,\\
N_1 &= 2W_0 W_1 - 4Z K_f V_1,\\
N &= N_0^2 - G N_1^2.
\end{aligned}$$

These equations are definitions of abbreviations. They do not add equations to (6) or unknowns to (7). An exact sparse-polynomial check independently compares this expansion with the conjugate product; the comparison has 295 monomials in its generic norm parameters.

2.3 The inherited arithmetic interface

We state precisely the reusable facts needed in the new proof. They are proved in the repository's ordinary blank-input, root-amalgamation and signed-domain notes, and preserved by the complete Wang coefficient adapter [4, 6, 9].

Proposition 2.1 (Weak transport and signed normalized interface)**.** *With the certified fixed tuple, strictly positive supplied coordinates, and $N_k = N_t = \epsilon \in \{1, -1\}$, one has*

$$C_w \geq 0, \quad q \geq 16, \quad 0 < R < q^4 - 2, \quad E \geq q^4 > R + 2.$$

If, in addition, integers u, τ, f, y, γ satisfy

$$\begin{aligned}
u &= X + ac + H\gamma, & u^2 - \Delta c^2 &= 1,\\
\tau^2 - \ell(\ell + k) &= 1, & f^2 - \Delta i^2 c^4 &= 1,\\
Q\big(c(Tf - 1) - Rf^2\big)^2 - (Q - 1)y^2 &= 1,
\end{aligned}$$

then $\epsilon = 1$ and a blank K215 halt can be recovered. The recovered main congruence coordinate and auxiliary roots may initially have signs. Conversely, a blank halt supplies canonical positive witnesses to the normalized interface, with all these quantities positive.

This is a substantive inherited theorem. In particular, it is not permissible to cite only its positive-root version after eliminating roots. Its proof uses signed Pell-rank congruences to identify the main index as R, excludes the negative outer sign and the boundary $C_w = 0$, recovers $X = 2^R$, and then decodes the native masks. The ordinary strong-equation formulation follows from the normalized one by $i_{\mathrm{ord}} = \Delta i$, preserving Q. The full inherited argument, including the signed-quotient issue, is retained with the archive rather than replaced by finite testing.

2.4 Coherent roots and a uniform conjugate bound

The first immediate strengthening of the weak bounds is

$$k = R + hE + \epsilon \geq R + E - 1 > 2R. \tag{9}$$

Here the positivity of the supplied h is essential. Also $X \geq 16$, $Y \geq 4096$, $a = Y(X+1)$, $c > 1$, $I = ic^2 > c$, and $Q > 1$.

Factoring (5) in a splitting field shows that $N = 0$ gives one coherent selection

$$u^2 = G, \quad \tau^2 = A, \quad f^2 = D, \quad dy^2 = Q(vf - b)^2 - 1, \quad S = Lu + J_\tau \tau + Hy. \tag{10}$$

Indeed, the full product is over the eight sign choices for u, τ, f of $d(S - Lu - J_\tau\tau)^2 - H^2 e + H^2 B_a f$; choose one vanishing factor and divide its linear expression by H to define y. Thus the signs are compatible by construction. No assertion about independence of square classes is used.

Lemma 2.2 (Smaller bound for every auxiliary conjugate)**.** *For every complex embedding of a splitting field containing* (10)*, its image y' satisfies*

$$|y'| < W\sqrt{G}, \qquad W = T^2 + XI^2.$$

Proof. Every embedding sends f to $\pm\sqrt{D}$. The auxiliary equation gives

$$|y'| < \sqrt{\frac{Q}{Q-1}}\,(v\sqrt{D} + b).$$

This also covers nonreal y'. The squared expression is real; its absolute value is bounded by $Q(v\sqrt{D} + b)^2 - 1$, which is positive and exceeds one. The ratio $I > c$ implies the exact inequality

$$\frac{D}{G} = \frac{1 + \Delta I^2}{1 + \Delta c^2} < \frac{I^2}{c^2}.$$

Together with $\sqrt{D} < (a+2)I$ and $\sqrt{G} > ac$, this gives

$$\frac{v\sqrt{D}}{\sqrt{G}} < TI, \qquad \frac{c}{\sqrt{G}} < \frac{1}{a}, \qquad \frac{D}{\sqrt{G}} < \frac{(a+2)I^2}{c}.$$

Since $\sqrt{Q/(Q-1)} < 3/2$, $R/k < 1/2$, and $c > kY$,

$$\begin{aligned} \frac{|y'|}{\sqrt{G}} &< \frac{3}{2}TI + \frac{3}{2a} + \frac{3}{2}\frac{R(a+2)I^2}{c} \\ &< \frac{3}{2}TI + \frac{3}{2a} + \frac{3}{4}(X+2)I^2. \end{aligned}$$

The last step uses $(a+2)/Y = X+1+2/Y < X+2$. Completing the square gives $\frac{3}{2}TI \le T^2 + \frac{9}{16}I^2$. Also $3/(2a) < I^2$. Consequently

$$\frac{|y'|}{\sqrt{G}} < T^2 + \left(\frac{3}{4}X + \frac{49}{16}\right) I^2 < T^2 + XI^2,$$

because $X \ge 16$. □

Lemma 2.3 (The two root separations)**.** *For every such conjugate y',*

$$J_\tau\sqrt{A} > L\sqrt{G} + H|y'|, \qquad L\sqrt{G} > H|y'|.$$

Proof. Since $\sqrt{G} < (a+2)c$, $\sqrt{A} > \ell = kXY^2$, and $c < k(Y+1)$,

$$\frac{\sqrt{G}}{\sqrt{A}} < \frac{X+2}{X}\frac{Y+1}{Y} \le \frac{9}{8}\frac{17}{16} = \frac{153}{128} < \frac{4}{3}.$$

Even $Y \ge 16$ suffices for this deliberately loose estimate. Since $HW > 1$,

$$4HW\sqrt{A} > 3HW\sqrt{G} > (1+2HW)\sqrt{G} = L\sqrt{G} + HW\sqrt{G}.$$

Lemma 2.2 proves the first assertion. The second follows from $L = 1+HW$ and the same bound. □

2.5 Rational roots become integer roots

If τ were irrational, an automorphism of a finite normal splitting field would send it to $-\tau$. Subtract the image of the last equation in (10) from the original. The magnitude of the changed τ term is $2J_\tau\sqrt{A}$, while the other changed terms have total magnitude at most $2L\sqrt{G} + 2HW\sqrt{G}$. This contradicts Lemma 2.3. Thus τ is rational and, since its square is an integer, integral.

If u were irrational, an automorphism sending u to $-u$ would fix the now rational τ. The same subtraction contradicts $L\sqrt{G} > HW\sqrt{G}$. Hence u is also an integer. The linear relation in (10) makes y rational. Moreover,

$$f = \frac{e - dy^2}{B_a}$$

is rational, since $B_a > 0$. Its integral square D makes f integral. It remains to justify integrality of y, which does not follow merely from $Hy \in \mathbb{Z}$.

Lemma 2.4 (Special auxiliary Pell saturation)**.** *If $m \ge 2$ and V are integers, $y \in \mathbb{Q}$, and*

$$m^2V^2 - (m^2-1)y^2 = 1,$$

then $y \in \mathbb{Z}$.

Proof. Write $y = a/b$ in lowest terms with $b > 0$. The equation gives $b^2 \mid m^2-1$. Set $D_0 = (m^2-1)/b^2$. It is a positive nonsquare integer: a square value would make m^2-1 a square. We have two positive integer Pell solutions, after taking absolute values:

$$m^2 - D_0b^2 = 1, \qquad (m|V|)^2 - D_0|a|^2 = 1.$$

Their ordinates cannot vanish. Write the positive powers of the fundamental integer Pell unit as $\chi_n + \psi_n\sqrt{D_0}$. For some $j, l \ge 1$,

$$m = \chi_j, \quad b = \psi_j, \quad m|V| = \chi_l, \quad |a| = \psi_l.$$

The addition formulas imply $\chi_{n+2j} \equiv -\chi_n \pmod{\chi_j}$ and $\chi_{2j-n} \equiv -\chi_n \pmod{\chi_j}$. Reducing l modulo $2j$ and reflecting into $0 \le n \le j$, strict increase of χ_n shows that $\chi_j \mid \chi_l$ is possible only for $n = j$. Thus l is an odd multiple of j. The Pell multiplication identities then give $\psi_j \mid \psi_l$. Hence $b \mid a$, and coprimality forces $b = 1$. □

Apply Lemma 2.4 with $m = \Delta I$ and $V = vf - b$. It makes y integral. Reducing the integral linear relation modulo H, and using $L \equiv 1 \pmod{H}$ and $J_\tau \equiv 0 \pmod{H}$, gives $u \equiv X + ac \pmod{H}$. Thus $\gamma = (u - X - ac)/H$ is an integer. All the equations of Proposition 2.1 are recovered, because $f^2 = D$ gives $vf - b = c(Tf - 1) - Rf^2$.

2.6 Soundness of the product and positive completeness

For any integer a, $\Delta = (a+2)^2 - 1$ is 0 or 3 modulo 4, and $G = 1 + \Delta c^2$ is 0 or 1 modulo 4. Therefore $N = N_0^2 - GN_1^2$ cannot be 2 modulo 4. At a zero of $P_{12} = N_k N_t (1 + N) - 1$, each of the three factors is an integer unit. The possibility $1 + N = -1$ would give $N = -2 \equiv 2 \pmod 4$, a contradiction. Consequently $N = 0$ and $N_k = N_t = \epsilon \in \{1, -1\}$.

The preceding reconstruction now applies without having assumed the signs it eventually proves. Proposition 2.1 forces $\epsilon = 1$ and recovers a blank K215 halt. In particular the product has no extra positive zeros arising from its unit signs.

Conversely, take a canonical positive normalized compiler witness. Let $u = X + ac + H\gamma$, and take its positive first and auxiliary roots τ, y. Define

$$r = \gamma + Wu + 4W\tau + y > 0.$$

Then $S = Lu + 4HW\tau + Hy$, so a coherent factor of the norm vanishes. The outer factors equal one. This constructs a positive zero of the system, the product, and the sum of squares.

Theorem 2.5 (Twelve-variable consistency representations)**.** *The system (6), the polynomial equation $P_{12} = 0$, and the polynomial equation $Q_{12} = 0$ have positive integer solutions exactly when K215 halts on its blank tape. Consequently ZFC proves that the unsolvability of each is equivalent to* Con(ZFC)*.*

2.7 Exact degree, circuit size, and the present degree obstruction

The important exact degrees are

Expression	c	a, H	Δ	I	D	Q, d	b	e
Degree	5	6	12	11	34	46	38	122
Expression	W	L, J_τ	V_0, A_0	V_1	C_0	C_1	N_0	N
Degree	24	30	82	41	134	87	536	1072

The term $H^2 e$ has degree 134, while $d(V_0 + A_0)$ has degree 128. Thus C_0 has unique leading term $-H_{\text{top}}^2 e_{\text{top}}$. The even coefficient N_0 has leading term $C_{0,\text{top}}^4$; the odd coefficient N_1 has degree at most 489, so GN_1^2 has degree at most 1000. It cannot cancel the degree-1072 term of N_0^2.

More explicitly, put $q_0 = \beta J$ and $k_0 = \eta + \zeta$. The norm's leading homogeneous form is

$$N_{\text{top}} = 2^{32} i^{48} k_0^{96} w^{80} s^{176} q_0^{656} (q_0 - F)^{16}. \tag{11}$$

It is nonzero for every fixed $\beta > 0$. Since

$$(N_k)_{\text{top}} = -hwsq_0^4, \qquad (N_t)_{\text{top}} = w(q_0 - F - \alpha) - zq_0,$$

the product has exact degree $1072 + 7 + 2 = 1081$. For example, the coefficient of

$$J^{677}hz\,i^{48}\eta^{96}w^{81}s^{177}$$

is $2^{32}\beta^{677} \neq 0$. The sum of squares has degree 2144.

The strong block is computed as $I = ic^2$, $I_2 = I^2$, $D_- = \Delta I_2$, $D = D_- + 1$, $Q = \Delta D_-$. It costs the same five gates as the old block, while exposing I^2 for reuse. Similarly $T_2 = T^2$, $v^2 = c^2T_2$ replaces $v = cT$, $v^2 = v \cdot v$ at the same cost; the actual norm circuit only uses v through v^2. This exposes T^2 for the weight. These shared computations recover the extra cost of the new first-root coefficient.

Presentation	Multiplications	Additions/subtractions	Total	Degree
Three-equation system	72	40	112	1072
Product	74	42	116	1081
Sum of squares	75	44	119	2144

Once the weights are this small, the dominant degree comes from H^2e, not from the separating weights. Merely reducing those weights further will not automatically lower this circuit's degree. A further substantial degree reduction should address the auxiliary constant $b = c + RD$, a different norm presentation, or a different arithmetic interface.

3 Nine natural unknowns with a smaller degree

The fewest-variable construction already present in package 18 [3] has nine nonnegative unknowns. We keep its explicit quartic-to-Jones specialization, replace its final relation-combining step, and use bounds adapted to the intermediate expressions. The resulting fixed polynomial has degree

$$\boxed{93264 \cdot 5^{33} + 288 = 10\,857\,358\,574\,867\,248\,535\,156\,250\,288}$$

and a supplied division-free circuit with 20 547 binary operations. The previous degree was $161844 \cdot 5^{33} + 484$, at 20 531 operations. Thus the degree decreases by approximately 42.3741%, for sixteen additional operations. The number of unknowns remains nine. This is an improvement of the supplied fixed-instance representation; it is not a new nine-variable representation theorem or a global minimality assertion.

3.1 The inherited quartic, specified completely

To make every coefficient determinate, we use the literal `quartic32.upstream.json` of package 18 [3]. In this subsection only, its positive coordinates are

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,\\ N, E, P, c, a, X, D, C_2, G, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta).$$

The four fixed ports are the entire Wang943 tuple $\kappa = (\beta, K_0, M_C, M_F)$; its fourth port is the shifted mask. The affine abbreviations are

$$q = \beta J + 1, \qquad k = \eta + \zeta, \qquad H = 4a + 3, \qquad C_w = q - F - \alpha.$$

The following 23 expressions are the residuals r_j:

j	r_j
1	$L - q^2 + 1$
2	$N - q(L + 1)$
3	$X - wq$
4	$a - E - sN$
5	$E - X(a - E)$
6	$c - k(a - E) - \eta$
7	$P - E(c - \eta) - k$
8	$\tau^2 - P(P - k) - 1$
9	$D - X - ac - \gamma H$
10	$\Delta - a^2 - H$
11	$C_2 - c^2$
12	$D^2 - \Delta C_2 - 1$
13	$k - jc + V - 1 - hE$
14	$G - (q - 1)(q - F) - \alpha$
15	$jc - V - LG - (M_C + qM_F)J$
16	$(K_0 + w)C_w + q - F - z(q - 1) - 1$
17	$F_2 - f^2$
18	$V - of + c$
19	$g - y^2 + V^2$
20	$I - iC_2$
21	$B_{\text{aux}} - I^2$
22	$B_{\text{aux}} - \Delta(F_2 - 1)$
23	$y^2 - B_{\text{aux}} g - 1$

All residuals have degree at most two in the 32 supplied coordinates. Define, with a dummy input x which does not occur,

$$\mathcal{P}_{32}(x, \mathbf{n}) = \sum_{j=1}^{23} r_j(\mathbf{n} + \mathbf{1})^2, \qquad \mathbf{n} \in \mathbb{N}^{32}.$$

It is a quartic: the coefficient of n_τ^4 is one. It satisfies the normalization $\mathcal{P}_{32}(x, \mathbf{0}) \geq 1$, because residual 5 equals one at the all-ones positive tuple.

The semantic dependency is the inherited ordinary-strong quartic theorem: these positive residual equations are equivalent in existence to the inputless native/Pell equations. That theorem is a four-port interface result [3, 9]. Package 19 [4] proves that its *entire* Wang943 tuple implements the same interface, including its canonical positive completion. Hence it can replace the original T4 tuple in all 23 residuals simultaneously. No conclusion is drawn from mask congruences alone, and no individual port is transferred in isolation. The ordinary compiler theorem and K215 calibration then give

$$\exists \mathbf{n} \in \mathbb{N}^{32}\ \mathcal{P}_{32}(1, \mathbf{n}) = 0 \quad \Longleftrightarrow \quad \neg\,\mathrm{Con}(\mathrm{ZFC}).$$

This remains an ordinary mathematical proof dependency, not a new proof-assistant endpoint.

The literal row 16 uses the old transport quotient. Some inherited notes write its shifted form $(K_0 + w)C_w = F + z_{\text{new}}(q - 1)$; the conventions are related by $z = z_{\text{new}} + 1$. Canonical witnesses have $z \geq 2$, as proved in the inherited positive-completion argument. The present coefficient recipe uses precisely the residual table above, without silently changing that convention. Inherited relative-path metadata inside the immutable source JSON is historical; the new complete binding is the Wang943 tuple specified in this report and its binding file.

3.2 Bounded definitions of every Jones coefficient

Put $\nu = 32$ and $L = 5^{33}$. For $\alpha = (\alpha_0, \dots, \alpha_{32}) \in \mathbb{N}^{33}$ with $|\alpha| \le 4$, define

$$p_\alpha = [x^{\alpha_0} n_1^{\alpha_1} \cdots n_{32}^{\alpha_{32}}]\mathcal{P}_{32},$$
$$c_\alpha = (4 - |\alpha|)! \prod_{j=0}^{32} \alpha_j!\, p_\alpha, \qquad e(\alpha) = \sum_{j=0}^{32} \alpha_j 5^j.$$

The factorial of the unused degree is part of the construction. There are $\binom{37}{4} = 66\,045$ eligible multi-indices. The delivered data record 346 nonzero coefficient polynomials in the four fixed ports.

After expanding each affine abbreviation, write

$$r_j(\mathbf{v}) = \sum_{a,b} d_{j,a,b} \mathbf{v}^a \kappa^b, \qquad d_{j,a,b} \in \mathbb{Z}.$$

The following is a finite, entirely specified integer recipe:

$$\rho_j = \sum_{a,b} |d_{j,a,b}|\, 2^{|a|} \kappa^b, \quad s_\star = 2 + 24 \sum_{j=1}^{23} \rho_j^2, \quad z_\star = 2^{s_\star}.$$

All four ports are positive. The coefficient ℓ^1 norm after replacing $\mathbf{v}$ by $\mathbf{n} + \mathbf{1}$ is at most ρ_j; this norm is submultiplicative, and every factorial multiplier is at most 24. Therefore

$$\sum_{|\alpha| \le 4} |c_\alpha| \le s_\star - 2, \qquad |c_\alpha| + 1 < z_\star.$$

Finally put

$$e_\star(T) = \sum_{|\alpha| \le 4} (z_\star + c_\alpha) T^{L - e(\alpha)},$$
$$u_\star = \sum_{j=1}^{32} (2z_\star)^{5^j}, \qquad y_\star = e_\star(2z_\star).$$

Since $e(\alpha) \le 4 \cdot 5^{32} < L$, every exponent is a fixed natural number. All digits are positive. These are the exact index constants in the constructive Jones specialization; $u_\star$ enters its proof interface but is not a leaf of the final circuit. There is no search for an unknown recursively enumerable index. The index integers are not materialized: their exact bounded recipes and modular evaluators are the delivered representation.

3.3 All nine-variable polynomial expressions

From here onward the symbols are local to the Jones construction. The nine supplied unknowns are exactly

$$(\varepsilon, g_0, h_0, s_0, w_0, \phi, i_0, j_0, n) \in \mathbb{N}^9.$$

Set $g = g_0 + 1$, $h = h_0 + 1$, $s = s_0 + 1$, $w = w_0 + 1$, $i = i_0 + 1$, and $j = j_0 + 1$. Define

$$b = y_\star + 1 + \varepsilon, \qquad \mathcal{B} = b^5, \qquad q = \mathcal{B}^L,$$
$$\ell = \sum_{r=1}^{32} \mathcal{B}^{5^r}, \qquad \lambda = \sum_{r=0}^{4L-1} \mathcal{B}^r, \qquad e = e_\star(\mathcal{B}),$$
$$\theta = \mathcal{B} - 2z_\star, \qquad N = q^{16}, \qquad G_{\mathrm{cap}} = 32 b \mathcal{B}^{5^{32}}.$$

The packed quantity is

$$\begin{aligned} R = {} & \left[g + eq^3 + \ell q^5 + \left(2(e - z_\star\lambda)(1 + \mathcal{B} + g)^4 + \lambda\mathcal{B} + \lambda\mathcal{B}q^4\right) q^7\right](N^2 - N) \\ & + \left[q^3 - b\ell + \ell + \theta\lambda q^3 + (\mathcal{B} - 2)q^8\right](N^2 - 1). \end{aligned}$$

Continue with

$$\begin{aligned} U &= N^2 w, & Y &= N^2 s, & M &= RY, \\ P &= 2M^2 U, & K &= R + 1 + h(P - 1), & A &= M(U + 1), \\ B_1 &= 2R + 1, & W_1 &= bw, & C &= B_1 + W_1 + \phi, \\ J &= 4A - 5, & Q &= 3W_1 C - 2(W_1^2 - 1), & D &= (A^2 - 1)C^2 + 1, \\ E &= iJDC^2, & F &= (A^2 - 1)E^2 + 1, & G_1 &= A + F(CD + 1 - A), \\ H &= B_1 + jC, & I &= (G_1^2 - 1)H^2 + 1. \end{aligned}$$

The two radicands, signed divisor, dividend, and margin are

$$\begin{aligned} a_0 &= (P^2 - 1)K^2 + 1, & a_1 &= DFI, \\ u &= JF, & v &= QF + J(H - C), \\ m &= K^2 G_{\mathrm{cap}} - 4(C - KY)^2 G_{\mathrm{cap}} - 8gK^2. \end{aligned}$$

All displayed quantities are polynomial abbreviations, not additional unknowns. In particular R is not assumed positive.

3.4 Unconditional bounds before imposing any equation

Lemma 3.1 (The packed expression never vanishes)**.** *On every nonnegative assignment of the eight outer coordinates, $R \neq 0$, $|A| \geq 2$, $P \geq 2$, and $K \geq 1$.*

Proof. The positive index recipe gives $b \geq 2$. Reducing the displayed formula for R modulo q gives

$$R \equiv (b - 1)\ell \pmod{q}.$$

Indeed $q \mid N$, and all terms in its second bracket except $(1 - b)\ell$ are divisible by q. Every exponent occurring in ℓ belongs to $\{0, \ldots, L - 1\}$, so

$$0 < (b - 1)\ell \leq (\mathcal{B} - 1)\ell \leq (\mathcal{B} - 1)\sum_{r=0}^{L-1} \mathcal{B}^r = q - 1 < q.$$

Thus $R \neq 0$. Put $\sigma = Y(U + 1)$. Since $U, Y \geq 1$, we have $\sigma \geq 2$ and $A = R\sigma$, whence $|A| \geq 2$. Moreover $P = 2R^2 Y^2 U \geq 2R^2 \geq 2$, and

$$K = R + 1 + h(P - 1) \geq R + P \geq 2R^2 + R \geq 1.$$

None of these conclusions uses a square condition, divisibility condition, or a sign inferred from m. □

Set

$$\begin{aligned} r &= W_1 + \phi, & \sigma &= Y(U+1), & T_C &= (j+2)C^2 + r^2, \\ M_D &= \sigma(R^2 + C^2), & M_F &= AE + 1, & & \\ M_I &= FDT_C + 2FH^2 + (F-1)A^2 + 1, & & & & \\ V_0 &= PK + 1, & V_1 &= M_D M_F M_I + 1. & & \end{aligned}$$

The circuit uses the equal, slightly cheaper expression $M_I = F(DT_C + 2H^2 + A^2) - A^2 + 1$.

Lemma 3.2 (Bounds adapted to the three Pell radicands)**.** *For every assignment under consideration,*

$$V_0, V_1 \geq 2, \qquad \sqrt{|a_0|} \leq V_0 - 1, \qquad \sqrt{|a_1|} \leq V_1 - 1.$$

Also $J \neq 0$, $D, F \geq 1$, and therefore $u = JF \neq 0$.

Proof. The preceding lemma gives $|A| \geq 2$, so $J = 4A - 5 \neq 0$ and $AJ = A(4A-5) > 0$. Consequently $D \geq 1$, $AE = iAJDC^2 \geq 0$, and $F \geq 1$. Since $K \geq 1$,

$$1 \leq a_0 = (PK)^2 - K^2 + 1 \leq (PK)^2,$$

which proves the V_0 assertion.

For D, substitute $A = R\sigma$ and calculate

$$M_D^2 - D = \sigma^2(R^4 + R^2C^2 + C^4) + C^2 - 1 \geq \sigma^2 R^4 - 1 \geq 0.$$

Also $M_D \geq 1$. Since $AE \geq 0$,

$$1 \leq F = (AE)^2 - E^2 + 1 \leq (AE+1)^2 = M_F^2, \qquad M_F \geq 1.$$

To bound I without an unstated sign hypothesis, use

$$H = (j+1)C - r, \qquad G_1 = FDC + F - A(F-1).$$

The elementary inequalities $|rC| \leq r^2 + C^2$, $|H| \leq H^2$ for integer H, and $|AH| \leq A^2 + H^2$ give

$$\begin{aligned} |CH| &\leq (j+1)C^2 + r|C| \leq T_C, \\ |G_1 H| &\leq FDT_C + FH^2 + (F-1)(A^2 + H^2). \end{aligned}$$

Thus $M_I \geq |G_1 H| + |H| + 1$, and

$$\sqrt{|I|} \leq \sqrt{G_1^2 H^2 + H^2 + 1} \leq |G_1 H| + |H| + 1 \leq M_I.$$

All three majorants are at least one, including when $C = 0$ or $H = 0$. Multiplying the bounds for $D, F, |I|$ proves $\sqrt{|DFI|} \leq M_D M_F M_I = V_1 - 1$. □

3.5 A signed-divisor relation-combining lemma

The second improvement avoids squaring both the divisor and dividend. The useful condition is a bound on the rational quotient, which is already available in this application.

Lemma 3.3 (Combining two squares, signed divisibility, and positivity)**.** *Let $a_0, a_1, u, v, m, M, V_0, V_1$ be integers satisfying*

$$u \neq 0, \quad V_0, V_1 \geq 2, \quad \sqrt{|a_t|} \leq V_t - 1 \ (t = 0, 1), \quad M \geq 1, \quad |v/u| \leq M - 1.$$

For $n \in \mathbb{N}$, put

$$\Omega = u(2m-1), \qquad \Theta = u(n+M) + v - \Omega(2M + V_0V_1)$$

and

$$\mathcal{C} = \left(\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)\right)^2 - 4\Omega^4 V_0^2 a_0 a_1.$$

Then

$$\exists n \in \mathbb{N}\, \mathcal{C} = 0 \quad \Longleftrightarrow \quad a_0, a_1 \text{ are integer squares}, \quad u \mid v, \quad m > 0.$$

Proof. Because $u \neq 0$ and $2m - 1 \neq 0$, also $\Omega \neq 0$. Choose complex square roots $r_t^2 = a_t$. Multiplication gives

$$\mathcal{C} = \prod_{\epsilon,\delta \in \{-1,1\}} (\Theta - \Omega(\epsilon r_0 + \delta V_0 r_1)).$$

If this vanishes, absorb the signs of a vanishing factor into the roots. Then $s = \Theta/\Omega = r_0 + V_0 r_1 \in \mathbb{Q}$. If $s \neq 0$, the identity $2sr_0 = s^2 + a_0 - V_0^2 a_1$ proves both roots rational, hence integral because their squares are integers. If $s = 0$, the bounds imply $|r_1| = |r_0|/V_0 \leq (V_0 - 1)/V_0 < 1$. The integer a_1 therefore has absolute value less than one, so $a_1 = a_0 = 0$. This also gives integral roots. No independence of quadratic fields is assumed.

Set $Z = 2M + V_0V_1 + r_0 + V_0 r_1$. The root bounds give

$$Z \geq 2M + V_0V_1 - (V_0 - 1) - V_0(V_1 - 1) = 2M + 1 > 0.$$

Dividing the vanishing linear factor by the possibly negative u yields

$$n + M + v/u = (2m-1)Z.$$

Its left side is at least $n + 1 > 0$. Hence $m > 0$; its right side is an integer, so v/u is an integer and $u \mid v$. Conversely, under the three stated conditions choose integer roots and put

$$n = (2m-1)Z - M - v/u.$$

This is an integer and $n \geq 2M + 1 - M - |v/u| \geq 2$. Its linear factor vanishes, proving the converse. This proves the lemma for both signs of u. □

In the Jones application choose the polynomial

$$M_{\text{div}} = Q^2 + (H - C)^2 + 1.$$

The identity

$$\frac{v}{u} = \frac{Q}{J} + \frac{H - C}{F}$$

and $J \neq 0$, $F \geq 1$ give $|v/u| \leq |Q| + |H - C| \leq M_{\text{div}} - 1$. No division occurs in the final polynomial; the rational identity is used only in its proof.

3.6 The resulting equation and consistency equivalence

Define

$$\Omega = JF(2m-1),$$
$$\Theta = JF(n+M_{\mathrm{div}}) + QF + J(H-C) - \Omega(2M_{\mathrm{div}} + V_0V_1).$$

$$\boxed{\mathcal{F}_9^{\mathrm{new}} = \left(\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)\right)^2 - 4\Omega^4 V_0^2 a_0 a_1.}$$

Every symbol on the right was defined by finite polynomial formulas. The requested equation in this section is $\mathcal{F}_9^{\mathrm{new}} = 0$ over $\mathbb{N}^9$.

Theorem 3.4 (The improved nine-variable specialization)**.** *With the exact coefficients fixed above, ZFC proves*

$$\left(\nexists \mathbf{x} \in \mathbb{N}^9 : \mathcal{F}_9^{\mathrm{new}}(\mathbf{x}) = 0\right) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

Proof. The new unconditional bounds and signed-divisor lemma show that, for each eight-coordinate outer assignment, existence of the ninth coordinate is equivalent to the same four Jones ratio conditions as in package 18: a_0, a_1 are squares, $JF \mid QF + J(H-C)$, and $m > 0$. The inherited constructive Jones lemmas `Nine.ratio_complete` and `Nine.ratio_sound`, composed with their packing reduction [12, 11, 3], identify existence of those eight coordinates with a nonnegative zero of $\mathcal{P}_{32}(1, \cdot)$. Our explicit normalized quartic and index recipes discharge their hypotheses exactly as in package 18. The ordinary quartic/compiler and K215 calibration then identify such a zero with a finite ZFC refutation. Negating the resulting existence equivalence proves the theorem. The change is in the finalizer and its bounds, not in the inherited ratio theorem. The full new endpoint has not been kernel checked. □

3.7 Exact degree and the operation ledger

All fixed ports and Jones index constants have degree zero. For a direct certificate, specialize every supplied unknown to t, keeping all fixed leaves symbolic. The following table gives exact degrees and leading coefficients on that line; the gate-by-gate calculation has no unresolved leading cancellation.

Expression	Degree	Leading coefficient
R	$235L$	1
P	$950L + 3$	2
K	$950L + 4$	2
A	$555L + 2$	1
a_0	$3800L + 14$	2^4
a_1	$24640L + 68$	2^{46}
JF	$6875L + 20$	2^{14}
m	$2221L + 11$	-2^9
M_D	$790L + 2$	5
M_F	$3160L + 9$	2^6
M_I	$8370L + 23$	2^{16}
V_0	$1900L + 7$	2^2
V_1	$12320L + 34$	$5 \cdot 2^{22}$
Ω	$9096L + 31$	-2^{24}
Θ	$23316L + 72$	$5 \cdot 2^{48}$

Here the two leading terms inside the outer square have equal degree, so one must check their coefficients instead of adding degree bounds blindly. Their difference has leading coefficient

$$(25-4)2^{96} = 21 \cdot 2^{96}.$$

The last term $4\Omega^4 V_0^2 a_0 a_1$ has lower degree. Therefore the final leading coefficient is $441 \cdot 2^{192} \neq 0$, independently of all fixed constants, proving the exact degree $93264L + 288$.

The delivered live circuit has 668 additions, 23 subtractions, and 19 856 multiplications, totaling 20 547 operations. It expands every fixed power into multiplication gates. It has exactly nine variable leaves and 348 nonliteral fixed leaves: $z_\star, y_\star$ and the 346 nonzero c_α. These are fixed integers with bounded recipes. Their bit lengths are not included in the operation count.

The independent verification reads the actual four Wang943 integers and checks their hashes, reconstructs the normalized quartic and all 346 coefficient records, and compares the literal DAG to direct finite sums modulo three primes. It also verifies the exact leading coefficient, 6,912 structured-majorant assignments including negative R, $C = 0$ and $H = 0$, and 134,400 bounded signed-finalizer evaluations, with 468 explicitly constructed completeness witnesses. These audits support implementation and transcription; they do not replace the universal proofs above or decide whether the polynomial has a solution.

4 One complete fixed coefficient tuple

4.1 The selected existing construction

Both new formulas use the complete compact Wang943 tuple from package 19 [4]. This is the smallest materialized base tuple in the inspected catalogue. Its base is

$$\rho = 2^{625}, \qquad d_0 = 5^{10} = 9\,765\,625, \qquad B = 2^{d_0} = \rho^{15625}.$$

The four ports are exactly

$$\beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_C, \qquad M_F = M_F^{\text{native}} + B - 1. \tag{12}$$

The last shift is indispensable. Substituting the unshifted native mask into a formula whose port is named M_F changes the polynomial and is not just a change of notation.

The complete tuple is reused through package 19's proved adapter for the inputless arithmetic interface. The common theorem requires the actual native language, carry bounds, anchor synchronization, population equation, and rank-adjustment property, not merely four integers with selected congruences. Reproducing those finite data checks supports the adapter; the ordinary soundness and completeness proof remains part of the cited mathematics.

4.2 A bounded recipe for all four integers

Here is a compact exact description. The ordered tile list `compiler/wang943_tiles.json` contains 943 quadruples of west, north, east and south edge colors. It is determined by the literal K215 table, with Start first, Forbidden second, and the remaining tiles in lexicographic order. The tile classes have counts

$$430 + 2 + 2 + 488 + 4 + 4 + 3 + 9 + 1 = 943 :$$

working transitions, halt loops, headless copies, directed head arrivals, finish tiles, initialization tiles, reset tiles, seam tiles, and the forbidden tile. Their full finite rule and the marked-periodic-tiling proof are preserved in `compiler/WANG943_COEFFICIENTS.md`.

Put $A_c = 1024$. For a tile $t_i = (x_{i0}, \dots, x_{i3})$, define

$$c_i = 1 + \sum_{s=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{is} + 512) A_c^{1+10s+j} + \sum_{j=41}^{44} A_c^j.$$

The extra bit encodes occupancy. The four ten-position copy groups have starting positions $1023, 1013, 1003, 983$, in west/north/east/south order; at copy position $e_s(j)$ put $c_{e_s(j)} = A_c^{1+10s+j}$. The dummy positions $1063, 1064$ have coefficient zero. At the four anchors $1095, 1096, 2195, 3296$, put $A_c^{41}, A_c^{42}, A_c^{43}, A_c^{44}$, respectively. Let $\mathcal{E}$ be this set of 989 selector, copy, dummy and anchor positions. Set

$$\mu = A_c - 2 + \sum_{j=1}^{44} (A_c - 3) A_c^j + 2^{450}(2^{78} - 1).$$

The compact layout is

$$\begin{aligned}
D_C &= \rho^{30} + \rho^{5517} + \rho^{1100} + \rho^{2200} + \sum_{e \in \mathcal{E}} c_e (\rho^{7631-e} + \rho^{8733-e}), \\
D_R &= \rho^{5497}, \\
M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\
M_F^{\mathrm{native}} &= 4 + \mu(\rho^{7631} + \rho^{8733}) + \sum_{j=0}^{9} \rho^{1013+j} + \sum_{j=0}^{9} \rho^{6520+j} + \rho^{2195} + \rho^{3296}.
\end{aligned}$$

All exponents are fixed nonnegative integers. These finite formulas and the fixed tile list completely determine (12).

4.3 Why the compact layout is compatible

The inherited adapter proves that the two clause bands have disjoint sets of possible rotation hits, so at least one is clean. The two anchors have only the intended simultaneous alignment. The tested targets have exactly their specified convolution contributions; unintended overlaps at untested positions are harmless. The largest unshifted convolution position is $12029 < 15625$, and the radix $\rho = 2^{625}$ exceeds the complete carry threshold. These facts are checked on the actual finite support, rather than inferred from stronger spacing conditions belonging to older layouts.

The population and residue invariants are

$$\begin{aligned}
\mathrm{popcount}(M_F^{\mathrm{native}}) &= 989, & \mathrm{popcount}(M_C) &= d_0 - 989, \\
M_C &\equiv 2 \pmod 4, & M_F^{\mathrm{native}} &\equiv 4 \pmod 8, \\
2D_C - D_R &\equiv 4 \pmod 5.
\end{aligned}$$

The last residue is a unit; it supplies the dummy adjustment needed for canonical rank congruences. Padding of the marked halting tableau supplies the prescribed positive witnesses. Because the complete arithmetic interface is preserved, the new root combination and the unchanged quartic parent may both use the tuple. This does not authorize combining one port from this tuple with a port from an older T4 tuple.

4.4 Materialized bytes and inherited attribution

We rebuilt the literal tile list, checked its source table, reproduced the compact geometry, and independently reconstructed every coefficient. The archive contains the four exact unsigned little-endian integers as deterministic gzip files. They decompress to:

Port	Bits	Raw bytes	Gzip bytes
β	9 765 625	1 220 704	1 219
K_0	13 201 251	1 650 157	16 111
M_C	9 765 625	1 220 704	1 616
M_F	9 765 626	1 220 704	1 314
Total		5 312 269	20 260

Each raw SHA-256 equals the published value in package 19. The full hashes are in `INSTANCE.json` and `compiler/COMPACT_COEFFICIENT_BINDING.json`. The literal K215 table has SHA-256

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090.`

The nine-variable coefficient recipe applies further finite operations to these same four integers. Its $z_\star, y_\star$ and 346 coefficient records are exact fixed constants, but they have not been expanded into binary integers. The 20,260-byte figure describes the four base ports only; it is not the expanded height or file size of every coefficient in F_9.

The compiler and machine data are reused work. Their source revision, original paths, license information, and copied proof notes are retained. The new contributions of this report are the algebraic weights, signed finalizer, resulting circuits and degree certificates, and their combination with this existing complete tuple.

5 Verification, reproduction, and integration

5.1 What was checked

For the twelve-variable construction, the independent verifier does not import either circuit emitter. It checks the complete generic norm identity coefficient by coefficient against a product of conjugate quartic factors. The comparison contains 295 monomials. It also verifies the 1649-monomial expansion of the uniform leading form (11), keeping all four fixed ports symbolic. Thus exact degree is certified for every valid positive β, rather than inferred from a few numerical substitutions. The final product and sum-of-squares tails are checked structurally. Twenty-four signed integer assignments and twenty-four rational assignments independently check the core formulas and shared-computation substitutions. The ledger validates every operand, topological order, and liveness.

For nine variables, the verifier regenerates the shifted quartic, checks all 346 weighted coefficient records, and authenticates the immutable parent JSON. A symbolic leading-term computation handles the actual cancellation inside the final norm and verifies the nonzero coefficient $441 \cdot 2^{192}$. Using the four materialized ports, it evaluates the exact bounded recipes for the index constants modulo three primes. Direct finite digit sums are compared with the generated DAG: twelve outputs and 312 named intermediates agree. The structured bounds and signed finalizer also receive the finite edge-case checks recorded in Section 3.

The coefficient reproduction rebuilds all 943 tiles from the literal 430 K215 actions, checks the helical control data, verifies the compact support geometry with 69,230 basis-convolution identities,

and independently reconstructs all four binaries byte for byte. The complete new mathematical proofs are reviewed separately. These finite checks detect implementation and transcription mistakes; they are not substitutes for quantified proofs.

5.2 Reproduction commands

From the extracted package directory, run

```
python3 reproduce.py
```

The default run uses only the Python standard library and the bundled gzip coefficients. It regenerates the two constructions in a temporary directory, compares their exact JSON with the delivered circuits, runs the independent verifiers, and writes the run receipt to `verification/REPRODUCED.json`. It does not require a GitHub connection or a full ProveIt checkout.

To rebuild and independently audit the coefficient bytes as well, run

```
python3 reproduce.py --rebuild-coefficients
```

This creates about 5.3 MB of raw coefficient data temporarily. No large Jones index integer is materialized. To compile the article using a normal TeX installation, run

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The included PDF was rendered and visually inspected after compilation.

5.3 Suggested integration

The archive is a self-contained package suitable for a new numbered subdirectory of the existing report. It includes a root README, the LaTeX article and compiled PDF, the three twelve-variable circuits, the nine-variable DAG and exact coefficient recipes, builders, independent verifiers, the complete compact coefficient binding and gzip payloads, mathematical review notes, and source provenance.

Existing reports should retain their identities. The new twelve-variable form supersedes the inspected twelve-variable products on degree and operation count together, while the old unrestricted-integer reductions, low-degree lifts, and 67-operation form remain relevant. The new nine-variable form is a further degree/operation tradeoff, not an improvement in variable count. Its immutable quartic source carries historical metadata, and the current binding is given explicitly in `INSTANCE.json`.

5.4 Formalization status

The K215 endpoint and the repository's parametric Jones reduction belong to existing formal developments. The new polynomial specialization, its new root bounds and finalizer, the complete native compiler, and the composed endpoint are not presented here as newly checked Lean or Rocq theorems. The delivered theorem is an ordinary proof formalizable in ZFC, using the identified inherited mathematical results. This distinction is preserved in every verification receipt and in the recommended README language.

6 Further research questions

1. Remove the new degree bottleneck. The weight reduction exposes the degree-134 term H^2e as the dominant part of the twelve-variable norm. Can an auxiliary-root translation, a different elimination order, or an ordinary rather than normalized strong equation reduce this term while retaining sound rational-to-integer recovery? Any proposed improvement must check zero denominators and all algebraic conjugates before assuming the desired roots are integers.

2. Transfer the new norm to fewer unrestricted integers. The present proof uses $h > 0$ to get $k > 2R$. The existing 26-integer lifts allow signed h and recover its sign later. Is there a different initial majorant or a staged reconstruction that permits the new low-degree norm on that larger domain? Substituting the old positivity encoding into this new proof without addressing that dependency would be unjustified.

3. Generalize the signed divisibility finalizer. Lemma 3.3 uses a polynomial majorant for $|v/u|$ to avoid squared divisibility. Can comparable quotient bounds replace squaring in other explicit relation-combining constructions, and which degree or circuit savings remain after the cost of the majorant is included? The lemma itself does not depend on the K215 compiler.

4. Improve the structured nine-variable majorants. The new V_1 has the same degree as $\sqrt{|a_1|}$, so a further major degree improvement cannot come from a lower-degree uniform polynomial majorant for unrestricted growth in the same variables. It may instead come from altering the underlying ratio representation, the finalizer, or the quartic parent. Can a lower-dimensional or lower-height quadratic parent reduce the fixed exponent 5^{33} while preserving a fully explicit coefficient recipe?

5. Mechanize the new lemmas and the interface boundary. The signed-divisor lemma has a short elementary proof, including its zero-sum radical case, and is a natural first formalization target. The generic norm identity and rational Pell saturation lemma are similarly reusable. A clean formal interface should expose the signed-root and weak outer-unit hypotheses explicitly, so that future algebraic reductions can be verified without reopening the native compiler proof.

6. Optimize coefficient geometry independently of algebra. The compact Wang tuple already gives concrete, portable base coefficients. Can the support and clause geometry be shortened further while preserving all rotation, carry, population and rank-adjustment properties? This is an independent finite optimization problem. A smaller verified whole tuple can then be substituted into both new formulas without changing their degrees or operation ledgers.

7. Maintain a reproducible comparison frontier. The present repository contains several genuinely different optimization objectives. A machine-readable catalogue with domains, exact-degree certificates, operation models, coefficient bindings and proof dependencies would make future claims easier to compare. In particular, comparison scripts should never silently count an abbreviation as a new unknown or count a large fixed exponent as one arithmetic gate.

References

[1] ProveIt contributors, *Diophantine forms for* Con(ZFC)*: current catalogue and comparison*, repository revision `ae3007ca4467011e0e051c84acb4a1dd13971b0e`. Pinned catalogue. All repository references below use this same revision, regardless of the older snapshot mentioned in a retained package's historical notice.

[2] ProveIt contributors, *17-primitive12*, package README.

[3] ProveIt contributors, *18-jones9*, package README, nine-variable emitter, and immutable quartic parent.

[4] ProveIt contributors, *19-wang943*, Wang943 language and coefficient proof, compact coefficient layout, and materialized payload manifest.

[5] ProveIt contributors, *20-root12*, root-combination proof and rational auxiliary Pell lemma.

[6] ProveIt contributors, *21-norm12*, norm proof, degree/operation tradeoff, and shared circuit emitter.

[7] ProveIt contributors, *22-monic12*, monic merger proof and retained-variable system.

[8] ProveIt contributors, *23-separating12*, twelve-variable proof and signed quotient absorption.

[9] ProveIt contributors, inherited ordinary and signed Pell/compiler arguments: ordinary blank proof, fixed blank product, signed-domain proof, and Pell interface audit. Copies are retained in `vendor/inherited/`.

[10] ProveIt contributors and attributed upstream authors, *K215 ordinary calibration and formal endpoint*: ordinary proof, literal transition table, and Lean endpoint. Retained provenance notices credit the numerical-calculus, machine, compiler, and formal-verification ancestry.

[11] ProveIt contributors, *Jones1982.Nine: nine unknowns*, parametric Lean endpoint, especially `wset_iff_nine`; the file imports the packing, ratio-condition and polynomial-representation developments.

[12] James P. Jones, Universal diophantine equation, *The Journal of Symbolic Logic* **47** (1982), no. 3, 549–571. doi:10.2307/2273588. The present specialization uses the repository's explicit implementation of the nine-unknown reduction and the source-level proof interfaces cited above.